# Algebra Monsoon 2026 Lecture 13 Abelianisation as a left adjoint

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## Abelianisation as a functor.

Let $G$ be a group.

A <u>commutator</u> in $G$ is an element of the form

$$(x,y) = xyx^{-1}y^{-1} \qquad (x,y \in G).$$

- $(x,y) = e \iff x$ and $y$ are permutable.

The <u>derived subgroup</u> $D(G)$ of $G$ is the subgroup of $G$ generated by all commutators in $G$.

- $G$ is commutative $\iff D(G) = \{e\}$.

- If $\varphi : G \to G'$ is a morphism of groups, then $\varphi((x,y)) = (\varphi(x), \varphi(y))$

Indeed $\varphi((x,y)) = \varphi(xyx^{-1}y^{-1})$

$= \varphi(x)\varphi(y)\varphi(x)^{-1}\varphi(y)^{-1} = (\varphi(x), \varphi(y))$ □

- $g(x,y)g^{-1} = (gxg^{-1}, gyg^{-1}))$

This is the special case $\varphi = Int(G)$ of the above claim.

- The quotient group $G/D(G)$ is commutative.

Ineed, if $\pi : G \longrightarrow G/D(G)$ denotes the surjective morphism, then

$$(\pi(x), \pi(y)) = \pi((x,y))$$

$$= e \text{ because } (x,y) \in D(G).$$

□

The quotient $G/D(G)$ is sometimes called the abelianisation of $G$.

## Universal property of $\pi : G \longrightarrow G/D(G)$

**Proposition** For every morphism $f : G \to A$ into a module $A$, there is a unique morphism of modules $G/D(G) \xrightarrow{\overline{f}} A$

such that $f = \bar{f} \circ \pi$.

Proof For every commutator $(x, y)$ in $G$, we have

$$f((x,y)) = (f(x), f(y))$$

$= e$ because $A$ is commutative.

Therefore $D(G) \subset \ker(f)$. By the universal property of the quotient $\pi : G \to G/D(G)$, there is a unique morphism of groups $\bar{f} : G/D(G) \to A$ such that $f = \bar{f} \circ \pi$. □

- Let $\alpha : G \to G'$ be a morphism of groups. There is a unique morphism of modules $F(\alpha) : G/D(G) \to G'/D(G')$ making the square

$$\begin{array}{ccc} G & \xrightarrow{\alpha} & G' \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi'} \\ G/D(G) & \longrightarrow & G'/D(G') \end{array}$$

$G/D(G)$ $\quad F(\alpha)$

commute.

Proof As $\pi' \circ \alpha$ is a morphism from $G$ into a module, there is a unique morphism of modules $F(\alpha)$ such that

$$\pi' \circ \alpha = F(\alpha) \circ \pi,$$

by the universal property of $G \to G/D(G)$.

□

For every group $G$, put $F(G) = G/D(G)$.

For a morphism of groups $\alpha : G \to G'$, let $F(\alpha) : F(G) \to F(G')$ be the unique morphism of modules such that the square

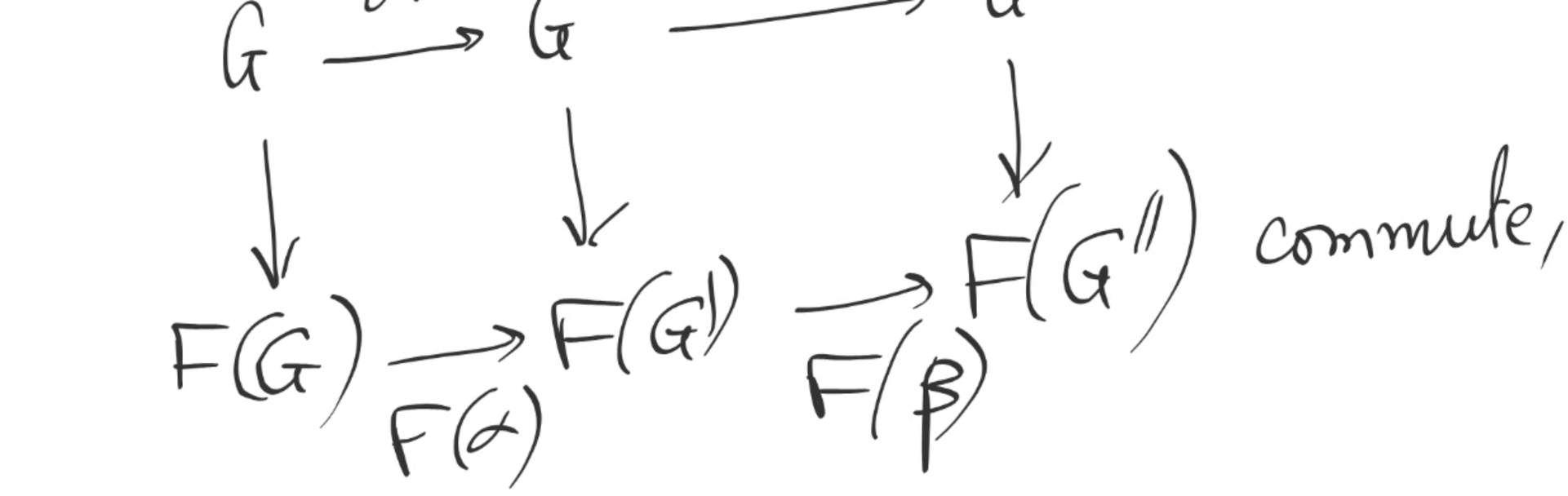
$$\begin{array}{ccc} G & \xrightarrow{\alpha} & G' \\ \downarrow & & \downarrow \\ F(G) & \xrightarrow{F(\alpha)} & F(G') \end{array}$$

commutes.

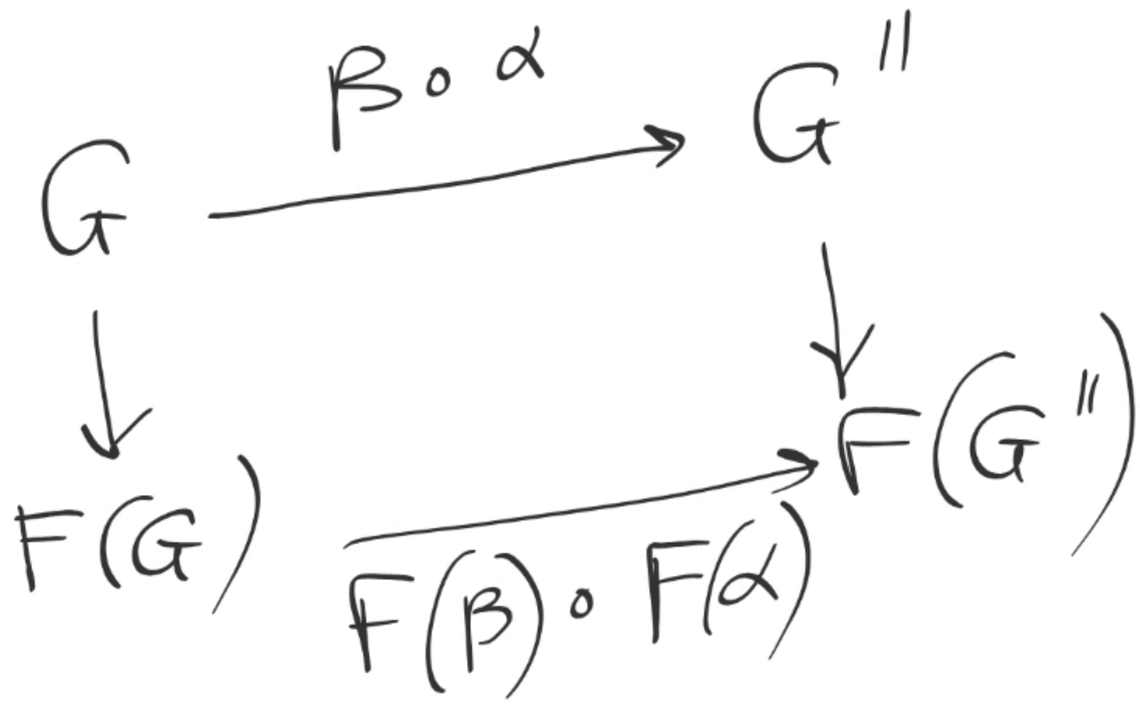
Proposition $F$ is a functor $\mathcal{G} \to \mathcal{A}$.

Proof It is clear that $F(Id_G) = Id_{F(G)}$.

If $\alpha : G \to G'$ and $\beta : G' \to G''$ then the squares




so




Commutes. By the uniqueness of $F(\beta \circ \alpha)$ we have $F(\beta \circ \alpha) = F(\beta) \circ F(\alpha)$. □

**Proposition** The functor $F : \mathcal{G} \to \mathcal{A}$ is a left adjoint of the inclusion $I : \mathcal{A} \to \mathcal{G}$.

**Proof.** Indeed, for every $f \in \mathrm{Hom}_{\mathcal{G}}(G, I(A))$, there is a unique $\overline{f} \in \mathrm{Hom}_{\mathcal{A}}(F(G), A)$ such that $f = \overline{f} \circ \pi$ (where $\pi : G \to F(G)$ is the canonical surjective morphism of groups).

The resulting bijection

$$\mathrm{Hom}_{\mathcal{G}}(F(G), A) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{A}}(G, I(A))$$

is functorial in $G$ and in $A$, so it is an adjunction.

check!

□